

## BREATHEBOX · DESIGN DECISION RECORD

# BreatheBox TRL 2 review decisions

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## DOCUMENT CONTROL

|             |                                                        |         |                |
|-------------|--------------------------------------------------------|---------|----------------|
| Document ID | BBX-DDR-001                                            | Version | 0.1            |
| Status      | <b>Draft</b>                                           | Date    | 2026-09-25     |
| Author      | Amish Chadha                                           | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                          |         |                |
| Repository  | github.com/BoujeeEnjinia1701/breathebox                |         |                |
| Source      | docs/decisions/0001-trl2-review-decisions.md @ 1cf85f1 |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                       |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Record the TRL 2 recommendations adopted for TRL 3 under Amish's 2026-09-25 instruction, open for his review, and the items that remain open |

## 0001: TRL 2 review decisions

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- **Date:** 2026-09-25
- **Status:** proposed. The recommendations in items A1 to A6 are adopted for TRL 3 work pending Amish's review; item O1 remains proposed, awaiting Amish.

### ■ Context

The TRL 2 review note ( docs/REVIEW.md , session of 2026-09-25) listed seven items as "Proposed, awaiting Amish", six of them with a recommendation. On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He did not review this repo's items one by one. The recommendations are therefore adopted as the basis for the TRL 3 calculations, model and drawing, and stay open for his review. Nothing in this record is decided or approved by Amish.

### ■ Options considered

The options for each item are in docs/REVIEW.md (TRL 2 session) and BBX-PRC-001. They are not repeated here.

### ■ Decision

*Table 1. Items adopted for TRL 3.*

| #  | Item                                                                                                                                  | Status                                                                                                                                                                                                                                                                                                | Where it now lives                                                       |
|----|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| A1 | Budget: recommendation (a), raise <code>budget_usd</code> from \$220 to \$250 and keep the bought counterflow core and the CO2 sensor | Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. The \$250 figure itself is awaiting Amish: <code>budget_usd</code> in <code>project.yaml</code> stays \$220, and BBX-CAL-001 states the cost against both figures. The priced BOM is now \$255, over both | BBX-CAL-001 section 9, BBX-REQ-001 R15, <code>bom/bom.csv</code>         |
| A2 | Window type for the first version: vertical sliding sash                                                                              | Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review. Casement and tilt-and-turn inserts are a later variant                                                                                                                                                    | BBX-PRC-001 v0.3, BBX-REQ-001 R8, <code>cad/src/model.py</code>          |
| A3 | Core type: bought polymer counterflow plate core (sensible only)                                                                      | Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review                                                                                                                                                                                                            | BBX-PRC-001 v0.3, BBX-CAL-001 section 2, <code>bom/bom.csv</code> line 2 |
| A4 | Frost strategy: slow the supply fan, with the combustion-appliance warning                                                            | Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review                                                                                                                                                                                                            | BBX-PRC-001 v0.3 (Safety), BBX-CAL-001 section 6                         |
| A5 | Nominal airflow: 50 m <sup>3</sup> /h with CO2-driven boost to 70 m <sup>3</sup> /h                                                   | Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review                                                                                                                                                                                                            | BBX-REQ-001 R1, BBX-CAL-001 sections 3 and 4                             |
| A6 | Controller: single ESP32-C3 class module with an SCD41 class sensor, with sensor checks on CalRig                                     | Adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review                                                                                                                                                                                                            | BBX-PRC-001 v0.3, <code>bom/bom.csv</code> line 7                        |

The TRL 2 review made no recommendation to change the pitch or problem wording, so `project.yaml` and `README.md` keep them.

## Items that remain open

Table 2. Open items.

| #  | Item                                                                                                     | Status                                                               |
|----|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| O1 | First co-design partner and region (for example a tenants' group or social landlord in the UK or Canada) | Proposed, awaiting Amish. No recommendation was made at TRL 2        |
| O2 | R6 noise shortfall found at TRL 3 (about 39 dB(A) at 50 m <sup>3</sup> /h against 30 dB(A))              | Proposed, awaiting Amish. Options are in <code>docs/REVIEW.md</code> |
| O3 | Budget figure: \$255 priced against \$220 in <code>project.yaml</code> and \$250 proposed                | Proposed, awaiting Amish                                             |

## Consequences

- The TRL 3 work is costed and sized for a sash-window unit with a bought counterflow core, two 24 V blowers, a CO2 sensor and a slowed-fan frost mode.
- BBX-CAL-001 shows R6 and R15 not met; neither is resolved by these adoptions.

- The CO2 sensor check depends on CalRig. CalRig's R9 (CO2 span 400 to 2,000 ppm against a reference of  $\pm(30 \text{ ppm} + 3 \%)$ ) covers the 1,000 ppm range BreatheBox needs, and an SCD41 class breakout fits CalRig's 90 x 70 x 50 mm bay (R10). This is consistent with `calrig/docs/REVIEW.md`.
- Amish may reverse any item; the calculations would then be rerun.